

Support-ticket MCP server

Answers questions about 40,047 support tickets, says which ones the data cannot answer, and finds that half of the archive's first replies ask for details without naming any.

A. DESIGN

Four MCP tools over stdio and no LLM in the server: the host supplies the model; the server, grounded facts and refusals. Corpus: **40,047** bilingual (en/de) tickets from Tobi-Bueck/customer-support-tickets, customer message, first reply, queue label. It is synthetic, built for classifier training.

- `get_schema`: columns, their values, where the rows came from (61,765 downloaded, 40,047 served), and an explicit `cannot_answer` list (no timestamps, customer ID, assignee or money), so the host refuses instead of confabulating.
- `run_sql`: one read-only SELECT against DuckDB.
- `find_similar_tickets`: multilingual-e5-small, mean-centred (unrelated pairs score ~0, not ~0.8); exact search over **31,625** train-split vectors. Below a **0.47** floor: `has_precedent`: false. Every hit carries `reply_state`.
- `suggest_routing`: k=1 nearest-neighbour vote (k swept on val; larger k floods small queues), with two confidence figures, top-1 similarity and top-3 agreement, fitted on validation and checked on test.

Fetch once; eight stages rebuild DuckDB and the index in ~80s, deterministic, offline after that. Every threshold and metric the tools state comes from the build's files. Nothing here governs whether the host calls a tool, only what it gets back.

B. WHY

Support over Churn because Churn is one numeric table and SQL over it is solved; tickets are text plus fields, so retrieval fits the text, SQL fits the fields, and the host chooses.

D. RISKS

An agent drafts a reply with no real precedent	Floor 0.47 , fitted on 400 validation tickets outside the index: accepts 97.8% of them and rejects the 20 off-topic queries it was fitted against, both in-sample; on 40 held-out typed queries it rejects 24 of 25 off-topic and accepts 6 of 15 on-topic. Below it, hits come back for context, not as draft material. Typed questions score ~0.18 lower whatever their topic, partly because they are shorter, so for them the floor is advisory.
A confidence figure is over-trusted	Both queue figures, fitted on validation and checked on test: in the upper bands similarity holds within 0.006, agreement runs 0.034 high, so the tool calls that one a ceiling; when they differ by >0.15 (half of test tickets) it names the lower as the conservative read. Priority has no confidence figure (macro-F1 0.766).
Source language label wrong on 10% of rows	4,204 tickets labelled <code>de</code> carry English text. <code>language</code> is re-detected from the text; the original stays as <code>language_label</code> .
Silently wrong tag aggregates	Positional <code>tag_1..8</code> make <code>GROUP BY tag_1</code> run clean and wrong; they are not served, and <code>ticket_tags</code> holds one row per (ticket, tag), so the wrong query is unwritable.
SQL reads files or mutates	Read-only, external access off, config locked, one SELECT via the parser, a cursor and limits per call. Capability removed, not names blacklisted. Residual: metadata functions reveal the database path.
Memorised test set inflates scores	8,357 tickets appear in both multi-language files; split first and ~28% of test has a twin in training, so dedup precedes the split and the index holds train only. Beyond those, 55% of test tickets have a paraphrase in the index; on the rest, macro-F1 is 0.414.
German quietly worse than English	0.782 en / 0.542 de macro-F1. Mostly coverage, not the encoder: retrieval is monolingual (96% of German queries' nearest neighbour is German) and the German index slice is 2.3× smaller; English cut to that size scores 0.545. Fix: more German tickets, not a bigger model.
Prompt injection via ticket text	Retrieved text is untrusted: the server only embeds and returns it, and the tool description tells the host to read it as data, not instructions. Authorisation would live server-side (C), never with the model.

E. BUSINESS IMPACT

Value is elapsed time, not agent minutes: a ticket settled in one exchange costs hours, one that needs three costs days, and the first reply decides which. "Send the system logs and your account number" can be acted on. "Please provide details" cannot.

49.8% of first replies in this archive are the second kind: they neither resolve the ticket nor name anything fetchable. (*Haiku 4.5 over 39,739 answers, \$9.53; on the 40 I read by hand, model and reader both count 22 dead ends.*) The archive is synthetic and the label a text proxy, so this is the pilot's hypothesis: a dead-end first reply costs an extra exchange, which thread length and resolution time on real tickets would test. The same labeller over CHEQ's

- **Avoided: an LLM server-side.** The host already has one; a second would double cost per call and make every tool non-deterministic and hard to evaluate. The one LLM pass, labelling 39,739 first replies as dead end or not (E), ran once and its output is committed.
- **Avoided: a fine-tuned classifier.** It would likely score higher but returns a softmax with no evidence; kNN returns neighbours a user can check, and their agreement is a second confidence signal.
- **Avoided: a reranker.** `subject + body` over-ranked subject-less tickets 2.6×; embedding the body alone removed it.

C. PRODUCTION

- The four tool contracts stay; everything beneath them changes.
- **Source.** CSVs → DuckDB becomes Zendesk/Jira → Snowflake. The index rebuilds on an Airflow schedule and the corpus-relative floor recalibrates with it.
 - **Transport.** stdio → streamable HTTP: same tool code, second endpoint; a `draft_reply` tool lives there, since a webhook has no host model.
 - **Access control.** None today. In production, SQL is covered by Snowflake row-access policies *if the HTTP endpoint opens the session with the caller's own SSO token*. The vector index is a copy of rows outside that boundary, so re-gate it: resolve the caller's scope, mask the index, then take top-k. Post-filtering leaks existence and guts top-k for narrow scopes.
 - **Scale.** Brute force: 0.4 ms at 32k vectors, under 10 ms to about a million; an approximate index adds an error inseparable from the embedding's own.

own first replies gives its rate in a day; `reply_state` on every precedent lets a draft copy the replies the label calls resolved or actionable, not the dead ends.

At CHEQ, two uses of one pattern. Support agents draft first replies from precedent over CHEQ's own archive (C, Source), with the dead-end rate before and after as the first business number. And the Data & AI team puts internal Snowflake data behind Claude for stakeholders (a CSM asking why an account's invalid-traffic share jumped) with this server as the template: refuse what the data can't answer, fit every threshold on held-out data, publish the confidence next to the label, so stakeholders can trust numbers nobody hand-checked.

Routing 0.707 macro-F1 over 10 queues against 0.045 for always answering the largest queue · Claude Opus 5 passes **22 of 22** expected-value and refusal checks through the tools, unsourced extra numbers listed (`reports/host_eval.md`) · every evaluation metric here is generated into `reports/`.